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Research Project: Flicker photophobia as an experience of inefficient coding

Purpose and Approach

We will use a same/different task to measure temporal frequency discrimination within two flickering hemifields. The participant will view stimuli through eyepieces showing two side-by-side, spatially uniform hemifields, one for each eye. Participants will fuse the images using central fixation marks. On each trial (3 sec duration), participants will view a pair of flickering stimuli and report whether the stimuli are the same or different. The stimuli are presented out of phase. The reference flicker is presented on one side and the test flicker on the other side. In some of the trials, the test stimulus is 0 dB from the reference, in which case choosing "same" is correct. In the rest of the trials, the test is >0 dB, in which case "different" is correct. The psychometric function adjusts the test flicker frequency to estimate a threshold and slope for deciding "different", and the false positive rate for this report when the test dB value is zero. The participant will indicate same/different with the up/down arrows on a keyboard or the front mounted up/down buttons of a standard game controller. Data will be collected at two contrast levels crossed with two light levels (0.5 and 3.5 neutral density filter; NDF). In half the trials, the test frequency will be lower than the reference, and the other half will be higher. We will test 7 log-spaced frequencies between 3 and 20 Hz. The goal is to investigate differences in frequency discrimination between high and low light levels and between controls and people with migraine and photophobia.

Subjects

We will collect a complete dataset from 15 healthy, headache free participants, and 15 participants with migraine and photophobia ages 18-40. Participants will meet these criteria:

<u>Controls:</u>

- Classified by the POEM (Kaiser et al., 2019; and as updated in the POEM analysis code as of the date of this pre-registration) to be "headache-free".
- Score of 4 or below on CHYPS (1st quartile from Price et al., 2025).

<u>Migraine:</u>

- Classified by the POEM (Kaiser et al., 2019; and as updated in the POEM analysis code as of the date of this pre-registration) to be "migraine with aura". (Any aura type accepted)
- Score of 17 or above on CHYPS (3rd quartile from Price et al., 2025).
- Any number of headaches per month is accepted

<u>Exclusion criteria for all subjects include:</u>

• An answer of "yes" to the question "Do you have ongoing symptoms from a recent concussion?"

• An answer of "yes" to the question "Are you currently receiving treatment for a disease of the eye?"

• Abnormal color vision as judged by the Cambridge Color Test [>= 10 protan (red), >= 10 deutan (green), or >= 15 tritan (blue)].
• A best corrected visual acuity (BCVA) below 20/25 for either eye.
• A history of generalized epilepsy.

Participants will perform a practice task prior to starting data collection. Poor performance on this practice task may lead the participant to being excluded from the study (see below under *Subject*).

It is possible that we will identify other exclusion criteria as we screen subjects for the study. If participants are excluded for reasons identified prior to their participation in data collection, we will simply add these reasons to the list of exclusion criteria and not consider this a deviation from the protocol.

Participants will be permitted to listen to music or podcasts/audiobooks played from their smart phone speaker during the testing session or through headphones, but they need to be able to hear the feedback from the testing computer.

Participant Preparation
At the start of a data collection session, participants will be acclimated to the experimental room. The room illumination will be set so that the luminance of the walls of the room at eye level is roughly equivalent to the luminance of the stimulus field. The operator will review the experimental procedure with the participant and adjust the apparatus and chair for subject comfort. Half of the participants will start with the high light level and half will start with the low light level (assignment will be interleaved).

Apparatus, stimulus, data collection
All the stimuli will be generated with a digital light synthesis engine (CombiLED) under computer control. The stimuli will be presented through custom-made eyepieces with a semicircular, uniform hemi-field of 27.5° diameter. The participant will view the stimulus with their natural pupil. Visual stimuli are tailored to an individual observer's age, taking their predicted lens density and predicted pupil size into account (which in turn is based upon age, luminance of the stimulus background, and stimulus area). The background level of the stimulus is adjusted by passing the light through a 0.5 log NDF (high light level) or an additional 3 log NDF (low light level). The achieved luminance of the stimulus background will vary slightly by subject, but will be roughly ~3000-4000 cd/m^2 (high) or ~10-40cd/m^2 (low). The chromatic content of the background may also vary slightly between observers as this shift is needed to achieve the desired contrast levels. Ultimately, the nominal, targeted contrast upon the post-receptoral mechanisms will be set to 10% low and 30% high on LMS.

Participants will complete 4 sessions consisting of 28 blocks each (2 high and 2 low light level sessions). High and low sessions will alternate, and every other participant will begin with low light. More than one session may take place on a single day as long as there is a break of at least 1 hour between sessions. No more than two sessions will

be completed in a single day. The code uses QUEST+ (with a maximum of ± 5 dB from the reference) to adjust the test frequency and estimate the threshold.

Each session is composed of 28 blocks with 20 same/different trials. Each block measures one reference frequency. On each trial, one side (left or right, randomized) will show the reference frequency and the other will show the test (out of phase). The participant will decide whether the two stimuli look the same or different. The reference flicker frequency within a trial is drawn from log spaced set between 3 and 20 Hz. The test frequency will be set following QUEST+ following a psychometric function defined by Lesmes et al., 2015 (doi.org/10.3389/fpsyg.2015.01070). For roughly half of the trials, the test frequency will be forced to be 0 dB (same). Participants will receive feedback after each trial, with either a positive "beep" if they were correct (hit or correct rejection), or a negative beep if they were incorrect (miss or false alarm). The participant will be paid a bonus of 5 cents for each correct response during a session.

Data analysis and primary hypothesis testing

The psychometric function yields three parameters for each measurement (false positive rate, frequency difference corresponding to d'=1 performance, and the steepness of the psychometric function). Data will be analyzed within-subject, grouped across sessions and blocks into each of the four stimulus conditions (high and low contrast level, crossed with high and low light level at each reference frequency). We will compare the threshold and slopes of the psychometric functions between conditions. We will test the following hypotheses:

- Thresholds are a constant dB value across frequencies for a given contrast following Weber's law.
- Thresholds are elevated for low contrast stimuli
- Thresholds are the same across light levels
- Thresholds and slopes differ between controls and participants with migraine, reflecting a left-shifted and lower discrimination sensitivity function, similar to detection sensitivity (Coleston et al., 1994 doi.org/10.1136/jnnp.57.10.1208).

Data Exclusion Criteria and handling errors in data collection

Device

If the testing apparatus fails temporarily (e.g., due to a software crash) and it is possible to resume testing, then we will discard the single block during which the failure occurred, repeat this block, and then continue to complete the session. If the failure occurs between sessions, then the study can continue as originally planned after recovering from the failure. There may be failures of the testing apparatus that cannot be resolved within a few minutes. If so, we will discard all the data collected during this session, and we will invite the subject to return to re-collect data for the session.

Subject

If a subject leaves the study prior to the completion of 4 sessions of data collection, we will discontinue their participation in the study and exclude all their collected data.

If a participant needs to briefly stop data collection during a session, we will treat that circumstance in the same manner as a brief equipment failure that interrupts collection of data during a block. If a participant needs to stop data collection during a session and is unable to continue we will discard all data from that session and attempt to re-collect the session on a subsequent day. If the subject discontinues a second session we will discontinue their participation in the study.

Participants will complete a practice task. On the first session, the participant will continue completing blocks until they complete one block with zero errors. If the participant still makes an error after 10 blocks, they are not eligible to participate in the study. If this is not the first session, the participant will complete one block as a refresher.

The operator may observe subject behavior during the session that prompts concern that the collected data will be of low quality (e.g., excessive movement and fidgeting or sleepiness). The operator will discuss these observations with the study team, and data for a subject may be excluded from the study on this basis, as long as the decision is made prior to examination of the psychophysical data.

If participants are discontinued, we will recruit additional subjects in an effort to meet our prespecified study numbers.